



tic-tac-toe.zip
Zip Archive

I am working on a Python Pygame Tic Tac Toe project for my Software Engineering Lab. I have cloned the assigned repository and I want to fix the existing code based on the requirements in the README.

First, please go through the project and understand how the current code works. Please do not rewrite the whole project or make unnecessary changes.

For now, I only want to fix the current problems with the game. The diagonal win is not being detected. Also, when

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Worked for 56s >

Done. I kept the existing project structure and changed only the game logic needed for the current problems.

What I fixed

Diagonal wins

- Added both diagonal combinations to `check_winner()` :
 - Top-left → center → bottom-right
 - Top-right → center → bottom-left

Win on the final move

- `check_round_end()` now checks for a winner **before** checking whether the board is full.
- So if the final empty cell completes a winning line, it correctly shows the winner instead of "Draw!" .

No moves after the game ends

- `handle_click()` now immediately ignores clicks when `round_over` is `True` .

- After a winning move or draw, the turn is no longer switched and the computer does not make another move.

Also fixed the existing occupied-cell issue

- I noticed the existing code could overwrite a symbol if the player clicked an already occupied cell.
- I added a small check for that, since it is directly related to valid move handling.

I did **not** add the other README features or redesign the project.

I also ran focused tests for diagonal wins, final-move wins, draws, post-game clicks, and occupied cells. They passed.

Fixed project

Download the fixed Tic-Tac-Toe project

You can extract this ZIP and replace your current project with it, or copy the modified `rules.py` and `game_engine.py` into your cloned repository. </>



The fixes are working properly after I tested the game. Now I want to implement the next feature from the README. Please add a scoreboard that keeps track of X wins, O wins, and draws. The scores should continue across multiple rounds. When I restart only the current round, the scoreboard should stay unchanged. There should also be a way to reset the entire match, which should clear all the scores. Please keep the previous fixes working and make only the changes needed for this feature. Do not add the remaining features yet.

Worked for 1m 7s >

Done. I added only the scoreboard/match-management feature while keeping the previous fixes intact.

What changed

- **Scoreboard displayed at the top**
 - X: 0
 - O: 0
 - Draws: 0
- **Scores persist across rounds**
 - X win → X score increases.
 - O win → O score increases.
 - Draw → Draw score increases.
- **R — New Round**
 - Clears the current board.
 - Starts a fresh round.
 - **Does not reset the scoreboard.**
- **M — Reset Match**
 - Clears the board.
 - Resets X wins, O wins, and draws to 0.
 - Starts a completely fresh match.
- **Previous fixes remain**
 - Both diagonal wins are detected.
 - A winning final move is counted as a win rather than a draw.
 - Moves are blocked after the round ends.
 - Occupied cells cannot be overwritten.

I also added protection so the same finished round cannot accidentally increment the scoreboard more than once.

Testing

I tested:

- X win counting
- O win counting
- Draw counting
- Multiple rounds retaining scores
- R preserving scores
- M clearing scores
- Final-move win detection
- No moves after game over
- No duplicate score counting

All focused tests passed.

Updated project

Download the Tic-Tac-Toe project with the scoreboard

The full-match reset is currently mapped to `M`, while `R` remains the current-round restart.



The scoreboard is working properly after testing. Now implement the remaining features from the README. Add a way to choose whether X or O starts the round. Since O is controlled by the computer, if O is selected as the starting player, the computer should automatically make the first move. Also add separate controls for restarting the current round and resetting the entire match. Restarting a round should keep the existing scoreboard, while resetting the match should clear the scoreboard. Please preserve all the fixes and scoreboard functionality that we already implemented. Make only the necessary changes and do not redesign the project.

Worked for 44s >

Implemented the remaining requested features while preserving the previous fixes and scoreboard.

Added

- **Choose starting player**
 - Press `x` → X starts the round.
 - Press `o` → O starts the round.
 - Since O is the computer, selecting O automatically makes the computer's first move.
- **Current-round restart**
 - Press `R` → starts a new round.
 - The selected starter is preserved.
 - The scoreboard is **not** changed.
 - If O is the starter, O automatically makes the first move again.

- **Entire-match reset**
 - Press `M` → clears X wins, O wins, and draws.
 - Starts a fresh round while retaining the currently selected starter.

Previous functionality preserved

- Both diagonal wins work.
- A winning final move is counted as a win rather than a draw.
- Moves are blocked after game over.
- Occupied cells cannot be overwritten.
- Scores cannot be counted twice for the same round.
- Scoreboard continues across rounds.

I ran focused tests for starter selection, automatic O moves, round restart, match reset, scoreboard persistence, diagonal wins, final-move wins, and post-game clicks. **All tests passed.**

Updated project

[Download the final Tic-Tac-Toe project](#)



ChatGPT can make mistakes. Check important info.